DHANUSH P

Electronics and Communication Engineering Student

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CAREER OBJECTIVE

Aspiring Electronics and Communication Engineer seeking opportunities in Software, Embedded Systems, and Core Electronics to apply programming, problem-solving, digital design, microcontroller, and hardware-software integration skills to develop efficient and scalable solutions.

EDUCATION

K.S.R Institute for Engineering and Technology **2023 – Present**
B.E. Electronics and Communication Engineering CGPA: 7.87 (Up to 5th Semester)

SRV Innovative Public School, Rasipuram **2021 – 2023**
Class XII Percentage: 70.4%

SVB Modern Higher Secondary School, Rasipuram **2020 – 2021**
Class X Percentage: 73.6%

TECHNICAL SKILLS

Programming: C, Java (Basics), Python (Basics)

Software & Tools: Arduino IDE, Git/GitHub, VS Code

Core Concepts: Data Structures Basics, Object-Oriented Programming Basics, Problem Solving, Debugging

Embedded & Hardware: ESP32, Microcontrollers, Sensor Interfacing, Serial Communication

Other: Digital Electronics, Verilog HDL, Basic Image Processing

PROJECTS

Vehicle Number Plate Detection using ESP32-CAM

- Developed a vehicle number plate detection prototype using ESP32-CAM and embedded programming.
- Programmed the ESP32-CAM using Arduino IDE to capture and process vehicle images.
- Implemented basic image processing and pattern-based detection techniques for identifying number plate regions.
- Integrated camera hardware with software to enable real-time image capture and processing.
- Worked with serial communication, debugging, and hardware-software integration during development.
- Tested the application under different lighting and image conditions and improved the detection process.

Vehicle Alcohol Detection System using MQ-3 Sensor

- Developed an alcohol detection application using C programming, MQ-3 sensor, and microcontroller.
- Programmed the microcontroller to continuously read and process sensor values.
- Implemented threshold-based decision logic to identify alcohol presence.
- Developed an alert mechanism using buzzer/indicator based on processed sensor data.
- Performed debugging and testing to improve sensor reading reliability and system response.
- Applied concepts of input processing, conditional logic, real-time monitoring, and embedded software development.

INTERNSHIP

NextGen Instrument **Java Full Stack Intern — 15 Days**

- Gained hands-on exposure to Java-based application development and full-stack development concepts.
- Learned the fundamentals of front-end and back-end application development.
- Practiced Java programming, basic database concepts, and application logic.
- Worked on understanding the software development workflow, debugging, and problem-solving.

Hailstorm Technology, Coimbatore **Intern – Electronics & Sensor Systems**

- Worked with sensors and electronic components.
- Built circuits and interfaced microcontrollers.
- Learned real-time embedded applications.
- Gained practical experience in hardware-software integration and troubleshooting.

CERTIFICATIONS

- RTL Design using Verilog HDL

- Functional Verification using SystemVerilog and UVM
- NCC 'B' Certificate – A Grade
- NCC 'C' Certificate – C Grade

STRENGTHS

- Strong interest in Software, Embedded Systems, and Core Electronics
- Quick learner with practical problem-solving approach
- Ability to work effectively in team environments
- Willingness to learn new technologies and programming concepts